

EXPERIMENTS

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COURSE CODE: **DSA0602 DATA HANDLING AND VISUALIZATION**

SET 8 – EMPLOYEE PERFORMANCE ANALYSIS

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

QUESTION 1

Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

AIM

To create a line chart showing the performance trend of employees based on their years of service.

PROCEDURE

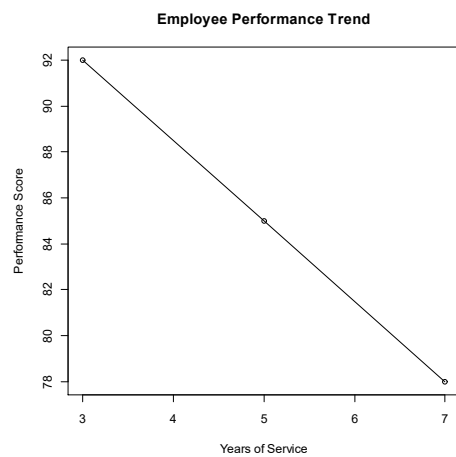
1. Create the employee performance dataset.
2. Store Employee ID, Department, Years of Service and Performance Score.
3. Use plot() to create a line chart.
4. Use Years of Service on the X-axis.
5. Use Performance Score on the Y-axis.
6. Add axis labels and a suitable title.

CODE

```
employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
employee <- employee[order(employee$Years_of_Service), ]
```

```
plot(  
  employee$Years_of_Service,  
  employee$Performance_Score,  
  type = "o",  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  main = "Employee Performance Trend"  
)
```

OUTPUT



RESULT

Thus, the line chart representing the employee performance trend was successfully created.

QUESTION 2

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

AIM

To create a bar chart showing the number of employees in each department.

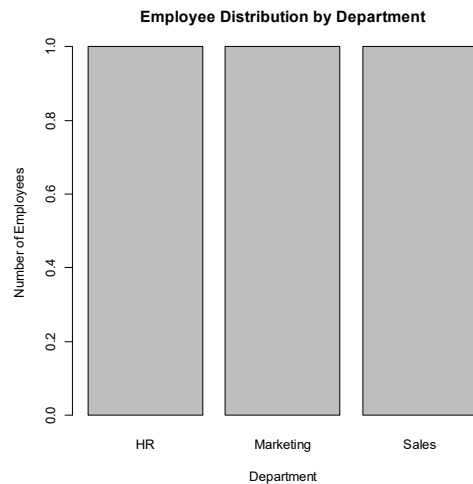
PROCEDURE

1. Create the employee performance dataset.
2. Count the employees in each department.
3. Use the table() function.
4. Use the barplot() function.
5. Add suitable axis labels and a title.

CODE

```
employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
department_count <- table(employee$Department)  
  
barplot(  
  department_count,  
  xlab = "Department",  
  ylab = "Number of Employees",  
  main = "Employee Distribution by Department"  
)
```

OUTPUT



RESULT

Thus, the bar chart showing the distribution of employees across departments was successfully created.

QUESTION 3

Build a table to display the performance data for each employee. Label the table elements.

AIM

To create and display a table containing the performance information of each employee.

PROCEDURE

1. Create the employee performance dataset.
2. Enter Employee ID, Department, Years of Service and Performance Score.
3. Assign suitable column names.
4. Display the data using print().

CODE

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
```

```
print(employee)
```

OUTPUT

```
Employee_ID Department Years_of_Service Performance_Score
1           1      Sales                5                85
2           2        HR                 3                92
3           3  Marketing                7                78
> |
```

RESULT

Thus, the employee performance data was successfully displayed in a labelled table.

QUESTION 4

Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

AIM

To develop an interactive employee performance dashboard using **R Shiny** by combining a line chart, bar chart and performance table.

PROCEDURE

1. Create the employee performance dataset.
2. Load the Shiny package.
3. Create the dashboard user interface.
4. Create the line chart for performance trend.
5. Create the bar chart for department distribution.
6. Create the employee performance table.
7. Add a department filter.
8. Run the Shiny application.
9. View the interactive dashboard in the browser.

CODE

```
library(shiny)
```

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
```

```
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
```

```

sidebarLayout(
  sidebarPanel(
    selectInput(
      "department",
      "Select Department:",
      choices = c("All", unique(employee$Department))
    )
  ),

  mainPanel(
    plotOutput("linePlot"),
    plotOutput("barPlot"),
    tableOutput("employeeTable")
  )
)

server <- function(input, output) {

  filteredData <- reactive({
    if (input$department == "All") {
      employee
    } else {
      employee[employee$Department == input$department, ]
    }
  })

  output$linePlot <- renderPlot({
    data <- filteredData()
    data <- data[order(data$Years_of_Service), ]

    plot(
      data$Years_of_Service,
      data$Performance_Score,
      type = "o",
      xlab = "Years of Service",
      ylab = "Performance Score",
      main = "Employee Performance Trend"
    )
  })

  output$barPlot <- renderPlot({
    data <- filteredData()

    department_count <- table(data$Department)

    barplot(
      department_count,
      xlab = "Department",

```

```

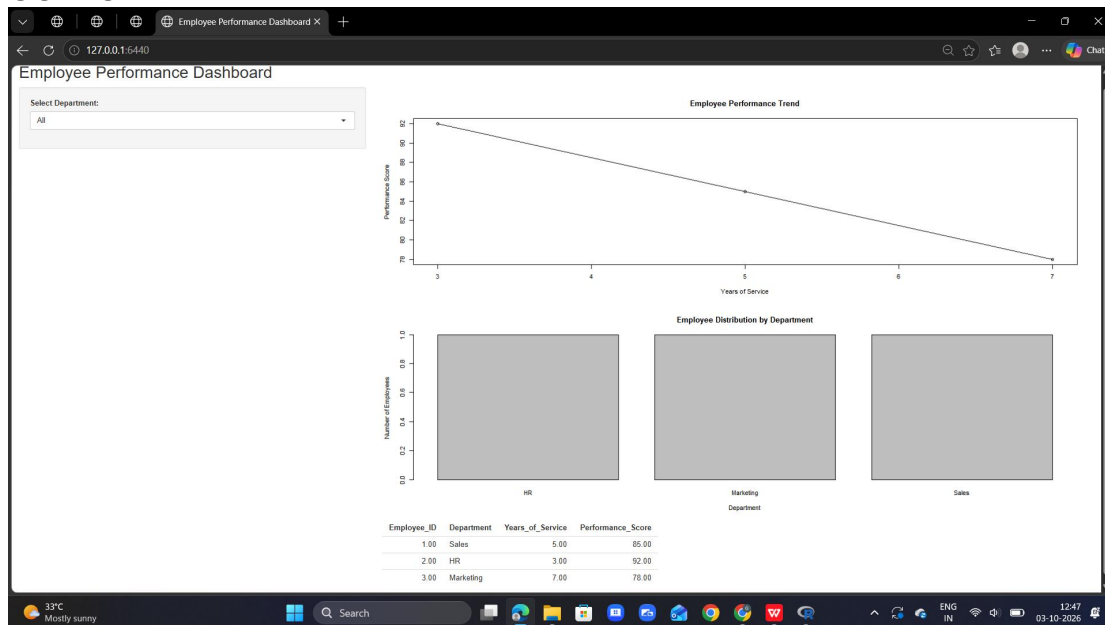
      ylab = "Number of Employees",
      main = "Employee Distribution by Department"
    )
  })

output$employeeTable <- renderTable({
  filteredData()
})
}

shinyApp(ui = ui, server = server)

```

OUTPUT



RESULT

Thus, an interactive employee performance dashboard containing the line chart, bar chart and employee performance table was successfully created using **R Shiny**.